MUHAMMAD HASSAN

Unreal Engine Developer (C++ | Blueprints) - Multiplayer (Replication/RPC) - Tools/Plugins - Pixel Streaming - Gaussian Splatting

Phone: +92 309 4367705
Email: tech34hassan@gmail.com
LinkedIn: http://www.linkedin.com/in/muhammed-hassan34
Portfolio: www.behance.net/muhammed-hassan

PROFESSIONAL SUMMARY

Unreal Engine developer with 4+ years building gameplay systems, multiplayer features, editor tools, and polished UI/UX in C++ and Blueprints. Proven experience delivering replicated multiplayer plus Pixel Streaming experiences, performance-first optimization, and custom pipelines for dense spatial capture (point clouds / splats) and cinematic automation.

CORE SKILLS

- **Unreal Engine 5.x**, C++, Blueprints, UMG, Editor Tools/Plugins
- **Multiplayer**: Replication, RPCs, Dedicated Servers, Epic Online Services (EOS), session flow (create/discover/join/leave)
- **Pixel Streaming** (browser-based gameplay, synced interactions)
- **Rendering/Tools**: RDG compute passes, GPU readback, export pipelines, logging/profiling/validation
- **3D Data**: Point grids/XYZRGB sampling, Gaussian Splatting workflows, Nerfstudio-compatible PLY export
- **Animation**: Animation Blueprints, locomotion, motion matching, Metahuman integration
- **Optimization**: Profiling/debugging, packaging, level streaming, memory/asset streaming
- GitHub, Perforce (P4)

PROFESSIONAL EXPERIENCE

Moshpit Studio - Game Developer | Jul 2021 - Present

- Shipped scalable **gameplay systems** (core mechanics, animation pipelines, modular architecture) designed for fast iteration and long-term maintainability.
- Delivered **replicated multiplayer** plus **Pixel Streaming** for smooth, synchronized browser-based gameplay.
- Built production-quality **UMG + C++/Blueprint tooling**, custom plugins, and documentation supporting reliable **PC + Android** releases.
- Drove **performance-first** profiling/debugging/optimization with cross-discipline collaboration and user testing.
- Built advanced **3D capture tooling** (LiDAR-style scanning, grid/route sampling, accurate mesh bounds) to generate dense, visibility-aware spatial datasets for point-cloud/splat workflows.
- Developed an **AI-driven cinematics system** that plans shots and auto-generates CineCamera actors, animation, and cuts, reducing manual keyframing effort.
- Implemented **data-driven editor UI generation** to auto-build/bind tool panels, reducing repetitive UI work and improving tool consistency.

Skylla Studio (Greece) - Multiplayer Developer | Apr 2025 - Present

- Built a multiplayer game on an Unreal Engine source build, integrated with **Epic Online Services (EOS)**.

- Implemented **dedicated server** architecture with session creation, discovery, and client join/leave flows.
- Developed networked gameplay features using **replication + RPCs** for smooth player interaction.
- Designed and implemented **UMG UI/UX** for menus and multiplayer flow screens.

Fiverr - Freelance Graphics & Animation Designer (Level 2 Seller) | May 2019 - Jun 2023

- Delivered **500+** design projects for Twitch/Discord creators and gaming communities.
- Produced motion graphics and creator branding assets (overlays, alerts, transitions, emotes, intros, logos) using After Effects, Photoshop, Illustrator, Premiere Pro.

PROJECTS

UnrealEngineTwin | May 2025 - Present

- Built an **EQS-driven autonomous navigation + scanning system** to generate dense, visibility-aware point grids and export **XYZRGB** samples for 3D reconstruction workflows.
- Implemented a real-time **UE5.5 Gaussian Splat capture + export** pipeline using **RDG compute passes + GPU readback**, with correct camera/world reconstruction and stable PIE capture.
- Created a **Nerfstudio-compatible PLY exporter** and added voxel + hierarchical variance-based merging to shrink outputs from hundreds of MB to tens of MB while preserving detail.
- Exposed tools via Blueprint-callable interfaces with strong logging, profiling, and validation for reliable large-scale exports.

Virtual Concert Experience (UE5 Technical Systems) | Aug 2022 - July 2024

- Built a full UE5 virtual concert platform with real-time show control, cinematic tooling, and scalable performance.
- Developed a **Nanite HISM crowd system** and optimized animation/material workflows in C++ + UE materials.
- Implemented runtime clothing switching and **HISM to Skeletal Mesh** conversion for interactive crowd characters.
- Created interactive lighting + fireworks sequencing with a custom timeline; added runtime camera recording + shot timeline and VR camera path recording.
- Integrated Pixel Streaming and optimized level streaming/packaging/memory/asset streaming to reduce load friction in large environments.

Multiplayer Squads Game (Invasion) | Mar 2025 - Present

- Built EOS-based multiplayer systems: sessions, squad creation, server browsing, achievements, and synchronized gameplay replication.

EDUCATION

BS, Information Technology - Punjab University College of Information Technology (PUCIT) | 2018 - 2022